

Lecture 16: Difference-in-differences

Economics 326 — Introduction to Econometrics II

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Motivation

- Previous lecture: treatment effects from **cross-sectional** data, assuming selection on observables (treatment is as good as random after controlling for covariates).
- Often hard to justify. Alternative: exploit **panel data** (repeated observations on the same units over time).
- **Difference-in-differences (DID)**: compare changes over time between a treatment group and a control group.

DID basic setup

- Two time periods: $t \in \{0, 1\}$ (before and after treatment).
- Two groups: $D_i \in \{0, 1\}$ (control and treatment).
- Treatment occurs between periods 0 and 1; only the treatment group ($D_i = 1$) is affected.
- Y_{it} : outcome for individual i at time t .

DID regression model

- DID regression:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

where $E[U_{it} | D_i] = 0$.

- Regressors:
 - t : time (0 = before, 1 = after).
 - D_i : group (0 = control, 1 = treatment).
 - $t \cdot D_i$: interaction, equals 1 only for the treatment group after treatment.

Interpreting the coefficients

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$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

- Evaluate $E[Y_{it} | D_i]$ for each combination of t and D_i :

$$\begin{aligned} t = 0, D_i = 0: & \quad E[Y_{i0} | D_i = 0] = \alpha, \\ t = 0, D_i = 1: & \quad E[Y_{i0} | D_i = 1] = \alpha + \gamma, \\ t = 1, D_i = 0: & \quad E[Y_{i1} | D_i = 0] = \alpha + \delta, \\ t = 1, D_i = 1: & \quad E[Y_{i1} | D_i = 1] = \alpha + \delta + \gamma + \beta. \end{aligned}$$

- Summarized as a 2×2 table:

	$D_i = 0$ (Control)	$D_i = 1$ (Treatment)
$t = 0$	α	$\alpha + \gamma$
$t = 1$	$\alpha + \delta$	$\alpha + \delta + \gamma + \beta$

- α : baseline (control group, $t = 0$).
- γ : pre-existing **group difference** at $t = 0$.
- δ : **time effect** — change in the control group from $t = 0$ to $t = 1$ (common trend).
- Change over time for each group:
 - Treatment**: $E[Y_{i1} | D_i = 1] - E[Y_{i0} | D_i = 1] = \delta + \beta$.
 - Control**: $E[Y_{i1} | D_i = 0] - E[Y_{i0} | D_i = 0] = \delta$.
- Subtract control's change from treatment's change:

$$(\delta + \beta) - \delta = \beta.$$

- DID estimand as a double difference:

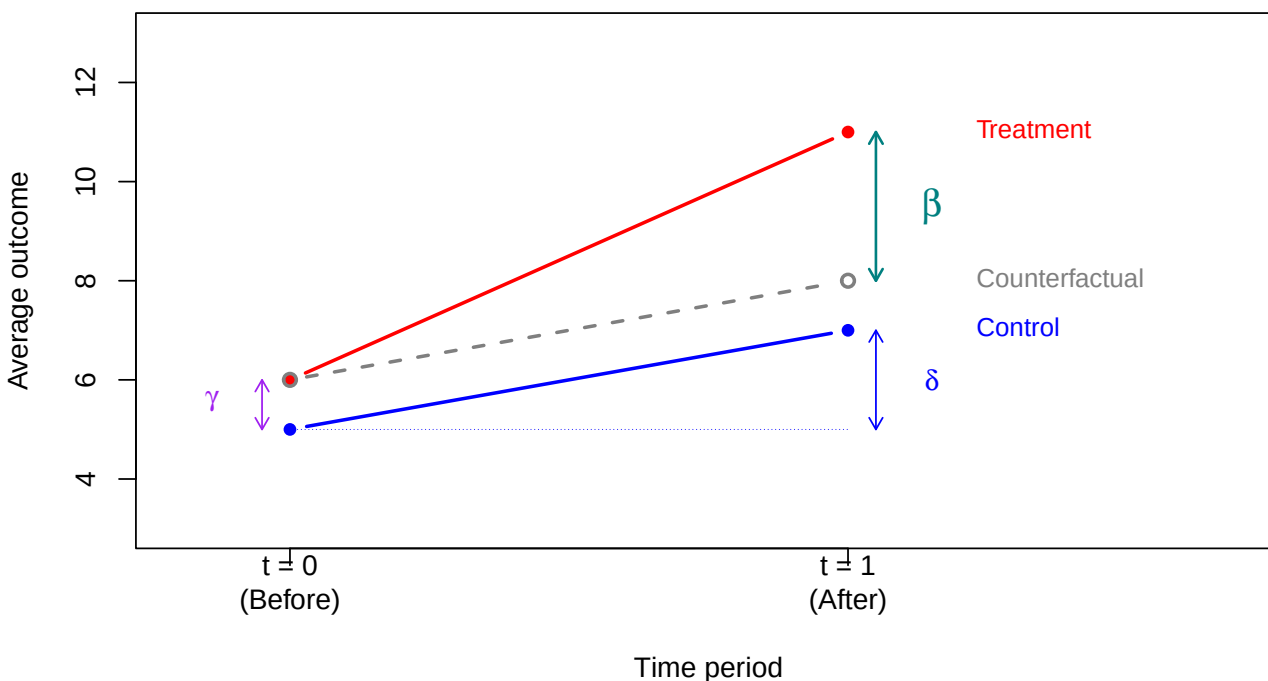
$$\beta = E[Y_{i1} - Y_{i0} | D_i = 1] - E[Y_{i1} - Y_{i0} | D_i = 0].$$

- The common trend δ cancels, isolating β .

DID diagram

- DID regression: $Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it}$.
- DID diagram — control, treatment, and counterfactual:

Difference-in-differences



- Counterfactual (dashed gray): treatment group's baseline $\alpha + \gamma$ plus the control group's change δ .

DID and potential outcomes

- **Panel potential outcomes:** $Y_{it}(d)$ — outcome for individual i at time t if assigned to group $d \in \{0, 1\}$.
- Four potential outcomes: $Y_{i0}(0), Y_{i1}(0), Y_{i0}(1), Y_{i1}(1)$.
- The observed outcome is:

$$Y_{it} = D_i Y_{it}(1) + (1 - D_i) Y_{it}(0).$$

- What we observe for each group:

	Control ($D_i = 0$)	Treatment ($D_i = 1$)
$t = 0$	$Y_{i0}(0)$	$Y_{i0}(1)$
$t = 1$	$Y_{i1}(0)$	$Y_{i1}(1)$

- Treatment effect on the treated at $t = 1$:

$$ATT = E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1].$$

The counterfactual $Y_{i1}(0)$ is unobserved for the treated group.

DID as a treatment effect

- $\beta = E[Y_{i1} - Y_{i0} \mid D_i = 1] - E[Y_{i1} - Y_{i0} \mid D_i = 0]$.
- Substituting observed outcomes with potential outcomes:

$$\begin{aligned} \beta = & E[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1] \\ & - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]. \end{aligned}$$

- To relate β to the ATT, add and subtract $Y_{i1}(0)$ and $Y_{i0}(0)$ inside the first expectation:

$$\begin{aligned} \beta = & E[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0] \\ = & E \left[Y_{i1}(1) - \underbrace{Y_{i1}(0) + Y_{i1}(0)}_{=0} - \underbrace{Y_{i0}(0) + Y_{i0}(0)}_{=0} - Y_{i0}(1) \mid D_i = 1 \right] \\ & - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]. \end{aligned}$$

- Rearranging and splitting the expectation:

$$\begin{aligned} \beta = & \underbrace{E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]}_{ATT} \\ & + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]}_{\text{difference in trends}} \\ & + \underbrace{E[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1]}_{\text{anticipation effect}}. \end{aligned}$$

- For β to equal the ATT, the difference in trends and anticipation effect must be zero. This requires two assumptions.

Assumption 1: Parallel trends

- Difference in trends is zero if both groups would have experienced the same change over time absent treatment:

$$E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0].$$

- Under parallel trends, the decomposition reduces to:

$$\beta = \text{ATT} + \underbrace{E[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1]}_{\text{anticipation effect}}.$$

- Cannot be directly tested: $Y_{i1}(0)$ is unobserved for the treated group.
- With multiple pre-treatment periods, can check whether trends were parallel before treatment.

Assumption 2: No anticipation

- Anticipation effect is zero if pre-treatment outcomes are not affected by future treatment assignment:

$$E[Y_{i0}(1) \mid D_i = 1] = E[Y_{i0}(0) \mid D_i = 1].$$

- Treatment assignment does not change pre-treatment outcomes in expectation.
- Under both parallel trends and no anticipation:

$$\beta = \text{ATT}.$$

Example: incinerator and house prices

- Kiel and McClain (1995): did a garbage incinerator in North Andover, MA reduce nearby house prices?
- Data: `kielmc` from the `wooldridge` package:

```
library(wooldridge)
data(kielmc)
# Show 2 observations from each of the 4 groups
rows <- c(
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 1), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 1), 2)
)
kielmc[rows, c("rprice", "y81", "nearinc", "age")]
```

	rprice	y81	nearinc	age
14	52000.00	0	0	32
15	49000.00	0	0	18
1	60000.00	0	1	48
2	40000.00	0	1	83
187	90245.77	1	0	1
188	46082.95	1	0	41
180	37634.41	1	1	81
181	39938.55	1	1	71

- `rprice`: house price in 1978 dollars (Y_{it}).

- `y81`: 1 if year is 1981 (after incinerator announced), 0 if 1978 ($t = 1$ if 1981, 0 if 1978).
- `nearinc`: 1 if house is near the incinerator site (D_i).

The 2×2 table of means

- Compute the four group means:

```
means <- tapply(kielmc$rprice, list(kielmc$y81, kielmc$nearinc), mean)
colnames(means) <- c("Far (nearinc=0)", "Near (nearinc=1)")
rownames(means) <- c("1978 (y81=0)", "1981 (y81=1)")
round(means, 2)
```

	Far (nearinc=0)	Near (nearinc=1)
1978 (y81=0)	82517.23	63692.86
1981 (y81=1)	101307.51	70619.24

- Computing the DID by hand:

```
diff_near <- means[2, 2] - means[1, 2]
diff_far <- means[2, 1] - means[1, 1]
DID <- diff_near - diff_far
cat("Change (near):", round(diff_near, 2), "\n")
```

Change (near): 6926.38

```
cat("Change (far): ", round(diff_far, 2), "\n")
```

Change (far): 18790.29

```
cat("DID: ", round(DID, 2), "\n")
```

DID: -11863.9

DID regression

- The DID regression, where $y81nrinc = y81 \times nearinc$ is the interaction term:

```
options(scipen = 999)
reg_did <- lm(rprice ~ y81 + nearinc + y81nrinc, data = kielmc)
round(summary(reg_did)$coefficients, 4)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	82517.23	2726.910	30.2603	0.0000
y81	18790.29	4050.065	4.6395	0.0000
nearinc	-18824.37	4875.322	-3.8612	0.0001
y81nrinc	-11863.90	7456.646	-1.5911	0.1126

- Coefficient on `y81nrinc` matches the DID from the 2×2 table.
- $\hat{\beta} = -\$11,864$ (SE = 7,457, $p = 0.113$): negative but not significant at 5%.

Assumptions in the incinerator example

- **Parallel trends**: without the incinerator, house prices near and far from the site would have followed the same trend over time.
- **No anticipation**: before the incinerator was announced, living near the future site did not affect house prices.

DID with covariates

- Basic DID requires parallel trends to hold unconditionally. If near houses are systematically different (e.g., older) and age affects price trends, the groups may follow different trajectories even without the incinerator.

- Omitting age biases δ , which biases β .
- Adding covariates: **parallel trends conditional on age**. Among houses of the same age, near and far would follow the same trend.

```
reg_did_cov <- lm(rprice ~ y81 + nearinc + y81nrinc + age + I(age^2),
                  data = kielmc)
round(summary(reg_did_cov)$coefficients, 4)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	89116.5354	2406.0511	37.0385	0.0000
y81	21321.0418	3443.6311	6.1914	0.0000
nearinc	9397.9359	4812.2218	1.9529	0.0517
y81nrinc	-21920.2700	6359.7454	-3.4467	0.0006
age	-1494.4240	131.8603	-11.3334	0.0000
I(age^2)	8.6913	0.8481	10.2476	0.0000

- With age controls: $\hat{\beta} = -\$21,920$ (SE = 6,360, $p = 0.0006$) — nearly twice as large, highly significant.
- The change suggests the basic DID was biased by age-driven trend differences.

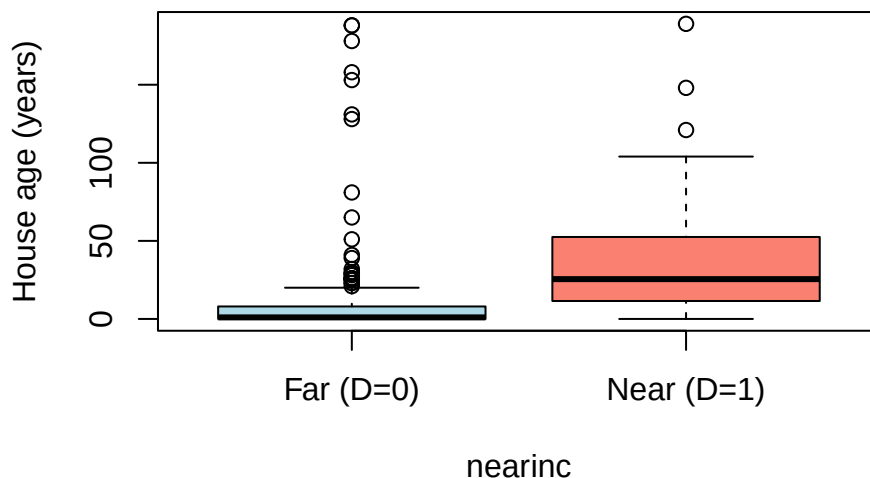
Where does the bias come from?

- Under no anticipation, the decomposition from the earlier slide gives:

$$\beta = \text{ATT} + \underbrace{\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]}_{\text{difference in trends}}.$$

- **No anticipation is plausible**: incinerator not publicly known in 1978.
- Bias must come from **difference in trends**: a **parallel trends** violation.
- Near houses are systematically older:

```
boxplot(age ~ nearinc, data = kielmc,
        names = c("Far (D=0)", "Near (D=1)"),
        ylab = "House age (years)",
        col = c("lightblue", "salmon"))
```



- Age affects price trends nonlinearly, so near and far groups would have had **different price trends even without the incinerator**.
- Estimated bias: $-\$11,864 - (-\$21,920) \approx +\$10,056$ — the age-driven trend difference partially masked the incinerator's negative effect.

Assumption	Violated?	Reasoning
No anticipation	No	Incinerator not yet announced in 1978
Parallel trends	Yes (unconditionally)	Near houses are older; older houses had different price trends

Detecting anticipation and pre-trends

- With only two time periods, anticipation effects **cannot** be separately identified.
- With **multiple pre-treatment periods**, use an **event study** design. Data span $t = -T, \dots, -1, 0, 1, \dots, T'$, where $t = 0$ is now the treatment date (not “before” as in the two-period model).
- Replace the single interaction $\beta(t \cdot D_i)$ with a separate coefficient per period. The baseline is $t = -1$ (last pre-treatment period):

$$Y_{it} = \alpha + \sum_{s \neq -1} \delta_s \cdot \mathbb{1}[t = s] + \gamma D_i + \sum_{s \neq -1} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

- Expected outcomes for $t = -2$ and $t = -1$:

	$D_i = 0$ (Control)	$D_i = 1$ (Treatment)
$t = -1$	α	$\alpha + \gamma$
$t = -2$	$\alpha + \delta_{-2}$	$\alpha + \delta_{-2} + \gamma + \beta_{-2}$
Backward trend ($t: -1 \rightarrow -2$)	δ_{-2}	$\delta_{-2} + \beta_{-2}$

- β_{-2} is the difference in backward trends from $t = -2$ to $t = -1$ between treatment and control: $(\delta_{-2} + \beta_{-2}) - \delta_{-2} = \beta_{-2}$.
- More generally, β_s for $s \leq -2$ is the difference in backward trends from $t = s$ to $t = -1$ between treatment and control.
- No anticipation + parallel pre-trends $\implies \beta_s = 0$ for all $s \leq -2$. However, we never observe $Y_{it}(0)$ for the treated group, so nonzero β_s tests the **joint** hypothesis: we cannot determine which assumption failed.
- The critical point: with multiple pre-treatment periods, we can **compare the two groups before treatment**. Since β_s for $s \leq -2$ are estimable from pre-treatment data, $\beta_s = 0$ is a **testable** implication of the joint hypothesis.
- β_s all equal and nonzero **suggests anticipation** at $t = -1$: backward trends are the same for all $s \leq -2$, but something shifts at the baseline period.
- β_s nonzero and unequal **suggests a parallel trends violation**: groups were already diverging before treatment.
- For $s \geq 0$: under parallel trends and no anticipation, β_s measures the **treatment effect** at period s relative to $t = -1$.

Time fixed effects

- In the two-period model, the term $\delta \cdot t$ creates two intercepts: α at $t = 0$ and $\alpha + \delta$ at $t = 1$.
- With multiple periods $t = -T, \dots, -1, 0, 1, \dots, T'$, we cannot use $\delta \cdot t$ because that forces a **linear** trend: the time effect at period s would be $\delta \cdot s$, with no flexibility.

- Instead, include a separate dummy for each period. The regression actually estimated is:

$$Y_{it} = \alpha + \dots + \delta_{-2} \cdot \mathbb{1}[t = -2] + \delta_0 \cdot \mathbb{1}[t = 0] + \delta_1 \cdot \mathbb{1}[t = 1] + \dots + \gamma D_i + \dots + U_{it}.$$

- The key property: $\mathbb{1}[t = s]$ equals 1 when $t = s$ and 0 otherwise. At $t = -1$ all dummies are zero (baseline). At any other t , exactly one dummy equals 1:

Period	Active dummy	Intercept
$t = -1$	none (baseline)	α
$t = 0$	$\mathbb{1}[t = 0] = 1$	$\alpha + \delta_0$
$t = 1$	$\mathbb{1}[t = 1] = 1$	$\alpha + \delta_1$

- Each period gets its own intercept. The compact notation $\sum_{s \neq -1} \delta_s \cdot \mathbb{1}[t = s]$ writes the time dummies as a sum. The event study model with all the dummies written out:

$$Y_{it} = \alpha + \dots + \delta_{-2} \mathbb{1}[t = -2] + \delta_0 \mathbb{1}[t = 0] + \delta_1 \mathbb{1}[t = 1] + \dots + \gamma D_i + \dots + \beta_{-2} D_i \mathbb{1}[t = -2] + \beta_0 D_i \mathbb{1}[t = 0] + \beta_1 D_i \mathbb{1}[t = 1] + \dots + U_{it}.$$

- At each t , exactly one time dummy survives (and at $t = -1$ none do). So $\alpha + \delta_t$ is the intercept at time t . Define $\lambda_t = \alpha + \delta_t$ (with $\delta_{-1} = 0$):

$$\dots, \quad \lambda_{-1} = \alpha, \quad \lambda_0 = \alpha + \delta_0, \quad \lambda_1 = \alpha + \delta_1, \quad \dots$$

The λ_t 's are called **time fixed effects**. This is the regression we actually run in OLS, with a dummy for each period:

$$Y_{it} = \dots + \lambda_{-1} \mathbb{1}[t = -1] + \lambda_0 \mathbb{1}[t = 0] + \lambda_1 \mathbb{1}[t = 1] + \dots + \gamma D_i + \dots + \beta_{-2} D_i \mathbb{1}[t = -2] + \beta_0 D_i \mathbb{1}[t = 0] + \beta_1 D_i \mathbb{1}[t = 1] + \dots + U_{it}.$$

- At each t , exactly one λ -dummy equals 1 and all others are zero:

$$\underbrace{\dots + \lambda_t \cdot 1 + \dots}_{\text{only } \lambda_t \text{ survives}} = \lambda_t.$$

The notation λ_t is shorthand for this — it represents whichever λ is active at time t :

$$Y_{it} = \lambda_t + \gamma D_i + \sum_{s \neq -1} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

Individual fixed effects

- The same idea can be applied to individuals. In the event study model, the intercept for individual i is $\alpha + \gamma D_i$. This allows only **two** values:

Group	Intercept
Control ($D_i = 0$)	α
Treatment ($D_i = 1$)	$\alpha + \gamma$

All individuals within the same group share the same baseline. But individuals may differ even within a group (e.g., houses near the incinerator differ in size, age, neighborhood quality).

- **Individual fixed effects** give each individual its own dummy, just like time fixed effects give each period its own dummy. The regression is:

$$Y_{it} = \alpha_1 \mathbb{1}[i=1] + \dots + \alpha_n \mathbb{1}[i=n] \\ + (\text{time and treatment terms}) + U_{it}.$$

Only the dummy for individual i is active, so the intercept is α_i . Using time fixed effects λ_t from the previous slide:

$$Y_{it} = \alpha_i + \lambda_t + \gamma D_i \\ + \sum_{s \neq -1} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

- Since D_i does **not change over time** for any individual, it is already captured by α_i . Including both α_i and γD_i would be **perfect multicollinearity**, so we drop γD_i :

$$Y_{it} = \alpha_i + \lambda_t \\ + \sum_{s \neq -1} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

- The interaction terms $\beta_s \cdot D_i \cdot \mathbb{1}[t = s]$ survive because they vary across **both** individuals and time.

Two-way fixed effects

- Combining individual and time fixed effects gives the **two-way fixed effects** (TWFE) model:

$$Y_{it} = \alpha_i + \lambda_t \\ + \sum_{s \neq -1} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it},$$

where α_i is the individual fixed effect and λ_t is the time fixed effect.

- This is the standard **event study** specification used in the literature for DID with multiple periods.